Machine Learning & Neural Networks for Neuroscience

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All the parts were solved by me. I do not have anyone in my pair

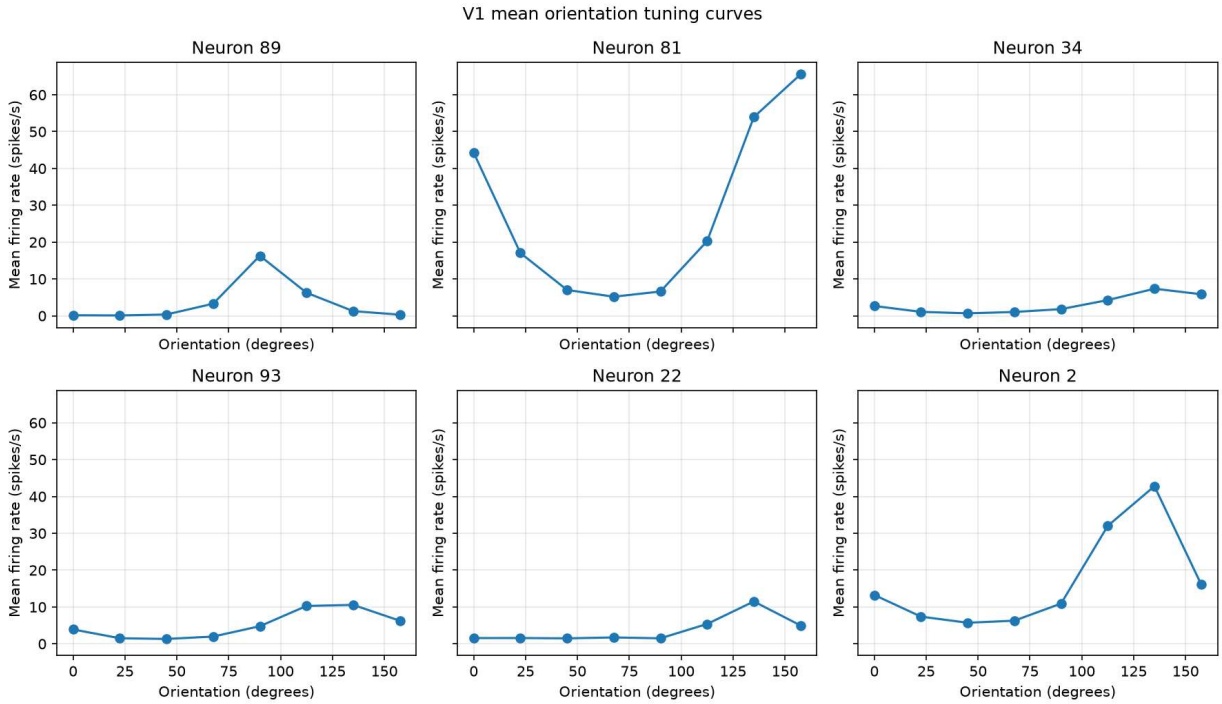
Part 1: From raw spikes to a tuning curve

Q1. How many neurons are recorded in the session you selected?
I've selected the session labelled '107l003p143_spikes.mat'. It has data recorded for 111 V1 neurons.

Q2. How many trials were presented for each orientation?
Each orientation has 400 trials.

Q3. What are the stimulus orientations?
0, 22.5, 45, 67.5, 90, 112.5, 180 and 157.5

Q4. Plot the mean orientation tuning curve for several neurons.
Orientation tuning curve plotted for six neurons



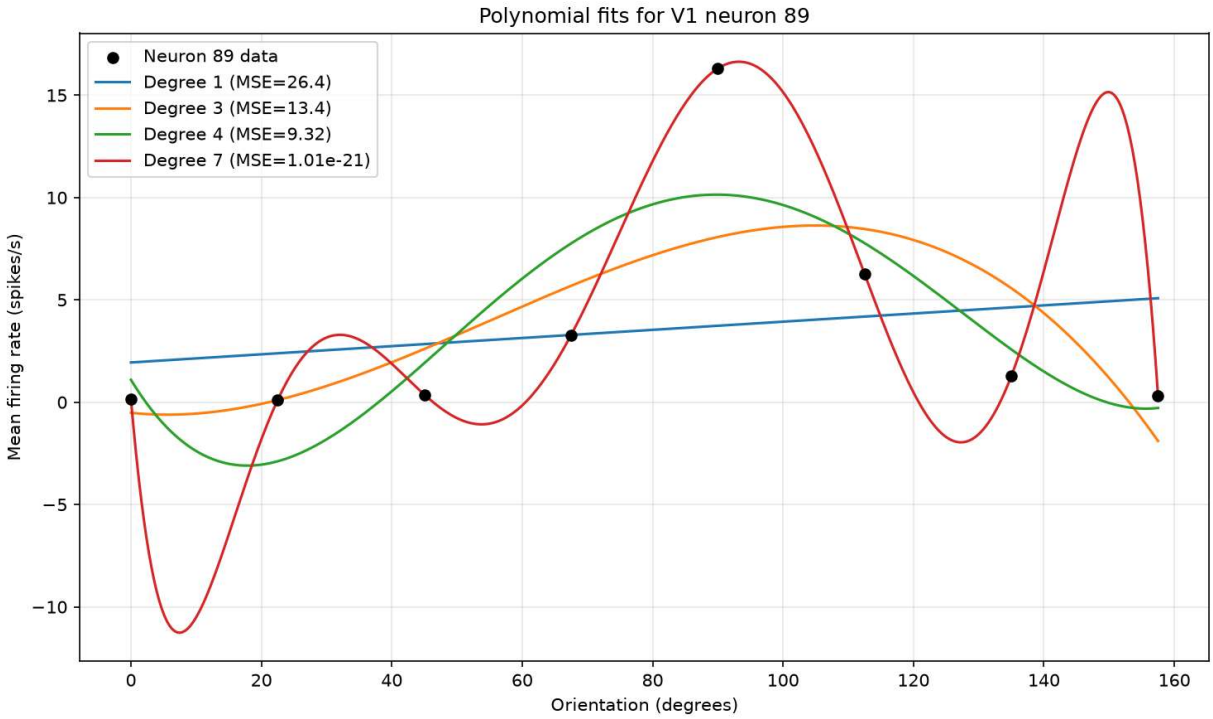
Q5. Select one neuron for the remainder of the assignment.
I've used a modulation score to select the neuron.

```
modulation_score = (
    (mean_responses.max(axis=1) - mean_responses.min(axis=1))
    /(mean_responses.max(axis=1)  mean_responses.min(axis=1))
)
```

The modulation score provides a normalized measure of response contrast, where values near zero indicate weak modulation and values near one indicate strong modulation. The normalization makes scores more comparable across neurons with different response magnitudes.
Based on the modulation score the neuron 89 (with a modulation score of 0.987) was selected. The score being close to 1 shows that the neuron's response changes strongly with orientation. It responds very weakly to some orientations and strongly to others.

Part 2: Polynomial regression

Plot the experimental data together with all three fitted functions.



Calculate the mean squared error (MSE) for each model.
Degree 1: MSE = 26.35949630; parameters = 2
Degree 3: MSE = 13.39568547; parameters = 4
Degree 4: MSE = 9.31937065; parameters = 5
Degree 7: MSE = 0.00000000; parameters = 8

Q1. Which polynomial degree provides the best fit?
Based on the training MSE, the seventh-order polynomial provides the best fit (since MSE ~0).

Q2. How does the fit change as the polynomial degree increases?
As the degree increases, the polynomial becomes more flexible, such that the linear model can only represent a straight-line relationship. The third-order model can represent a curved relationship with more changes in slope. The fourth-order model can represent more complex curvature. And the seventh-order model can closely pass through all eight measured responses.

Q3. What does the polynomial degree control?

The polynomial degree controls the complexity and flexibility of the model and increasing the degree allows the model to represent more changes in curvature.

Q4. What are the advantages and disadvantages of using a higher-degree polynomial?

Advantages

- Can represent more complicated response curves.
- Can achieve a lower training error.

Disadvantages

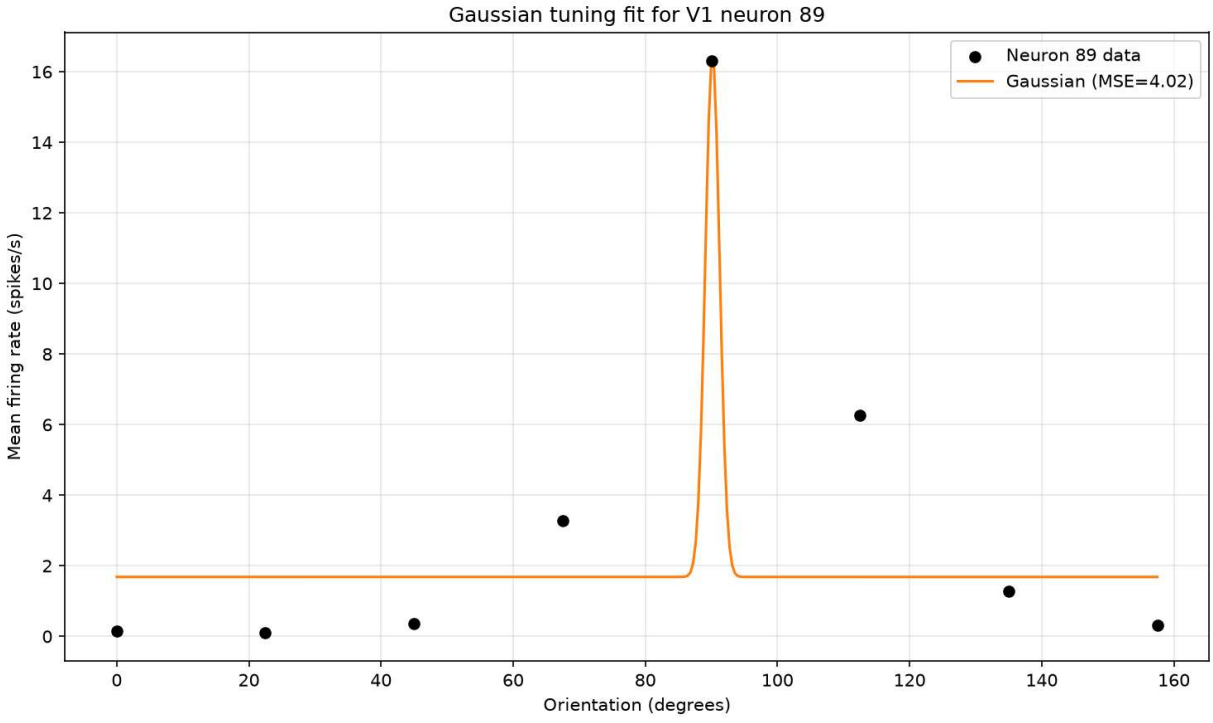
- Can overfit the measured data.
- Can oscillate strongly between data points.
- Predictions outside the observed range can become unrealistic.
- The fitted curve can be unstable if the dataset changes slightly.

Q5. Can the polynomial coefficients themselves be interpreted biologically?

No. Polynomial coefficients are useful for describing the mathematical fit, but they do not have a direct biological interpretation.

Part 3: Nonlinear regression

Fit the nonlinear function to your neuron's data.



Calculate its MSE and compare it with the polynomial models.

The MSE for the Gaussian model was 4.02013724.

Q1. Which model has the lowest MSE?
Among all fitted models, the seventh-order polynomial has the lowest training MSE (~0).

Q2. What is the estimated preferred orientation μ?
The preferred orientation is 90.16393753. This agrees with the raw tuning curve, which has its largest measured response at 90°.

Q3. What does σ tell you about the neuron's tuning?
The fitted width is 1.08481629.
A smaller σ means a narrower and more selective tuning curve. A larger σ would indicate that the neuron responds strongly over a wider range of orientations. For this fit, the small value of σ reflects the sharp peak near 90°. Since the orientations were sampled relatively sparsely, this fitted width should not be interpreted as highly precise without additional orientations near the peak

Q4. Which parameters of this model have a biological interpretation?
All four Gaussian parameters have useful interpretations:

- B: baseline response when the stimulus is far from the preferred orientation.
- A: response amplitude above baseline.
- μ: preferred orientation or stimulus label.
- σ: tuning width, describing how broadly the neuron responds.

Q5. Why might this nonlinear model be preferable to a polynomial even if their MSEs are similar?
The Gaussian model may be preferable because:

- Its parameters have direct biological meaning.
- It gives a clear measure of tuning width.
- It provides an interpretable baseline and response amplitude.
- Its shape is consistent with a localized orientation-tuning response.

A polynomial may have a lower training MSE but still produce biologically implausible oscillations and poor predictions at unmeasured orientations.

Part 4: A problem with orientation

Q1. Explain why this creates a problem for the Gaussian model used in Part 3.
The Gaussian model treats orientation as a linear variable. This means that it considers distance only on the ordinary number line. It does not recognize that the ends of the interval meet. For example, if the preferred orientation is near 0°, the Gaussian considers 170° far away, even though 170° is only 10° away in orientation space.

Q2. Why would a Gaussian centered near 0°have difficulty representing a tuning curve that extends across the 0°/180° boundary?
A Gaussian centered at 0° decreases as θ increases. It cannot automatically produce another high-response region near 180°. A circular tuning curve centered at 0° should be high at both 0° and

180°. A standard Gaussian on the interval [0°, 180°) instead treats these as opposite ends of a line. It therefore produces an artificial discontinuity at the boundary.

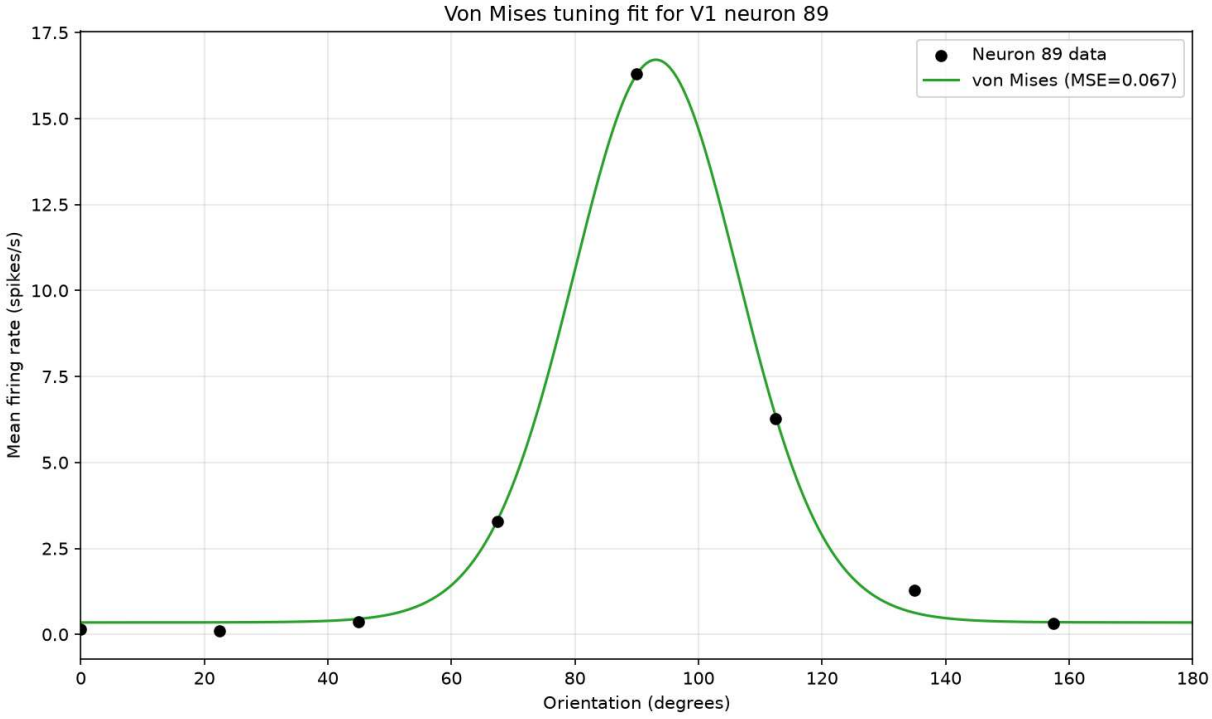
Q3. Why is polynomial regression also problematic for a circular variable?

Polynomial regression also treats orientation as a linear number. It does not enforce the identity that 0° = 180°. A high-degree polynomial can behave very differently at the two ends of the orientation range. It may also oscillate strongly near the boundaries and between measured orientations. Additionally, the polynomial coefficients depend on the arbitrary choice of where the circular coordinate begins.

Q4. How would you expect this problem to affect the neuron you selected?

Neuron 89 has a preferred orientation near 90°, which is far from the 0°/180° boundary. Therefore, the circularity problem is less severe for this particular neuron than it would be for a neuron preferring 0°. However, the orientation space is still circular, so a circular model is more appropriate in principle. The von Mises model also provides a better fit for this neuron.

Plot the experimental data together with the fitted circular model.



Calculate its MSE.

The MSE for the von Mises model is 0.06701762

Q1. How does the von Mises fit compare with the Gaussian fit?

The von Mises model (MSE=0.06701762) fits much better than the Gaussian model (MSE=4.02013724). The von Mises MSE is approximately 60 times smaller than the Gaussian MSE. It also gives a smooth tuning curve that respects the circular structure of orientation.

Q2. What is the estimated preferred orientation?

The estimated preferred orientation is 93.09598718°. This is close to the observed maximum at 90°.

Q3. What does the parameter κ tell you about the neuron's tuning?

The fitted concentration parameter is 4.54997226. Higher κ means that the response is more concentrated around the preferred orientation, corresponding to sharper tuning. Lower κ means broader and less selective tuning. The value for this model indicates a relatively concentrated orientation preference.

Q4. Why does the equation contain $2(\theta-\mu)$ rather than simply $(\theta-\mu)$?

Orientation has a period of 180° and not 360°. The factor of 2 ensures that the model has a period of 180°. Without the factor of 2, the model would have a 360° period and would treat direction as different when the assignment considers the orientations equivalent.

The factor of 2 also creates equivalent tuning peaks separated by 180°, which is appropriate for orientation rather than movement direction.

Q5. Does accounting for circularity improve the fit for your selected neuron?

Yes. Comparing the MSE values for the Gaussian and von Mises model shows that incorporating circularity substantially improves the fit for neuron 89.

Which model would you choose to describe this neuron's orientation tuning, and why?

For neuron 89, the Von Mises model is the best choice. The reasons are:

- Orientation is inherently circular.
- The model respects the equivalence of 0° and 180°
- The preferred orientation is interpretable.
- The concentration (κ) describes tuning sharpness.
- The model avoids unrealistic boundary behavior.

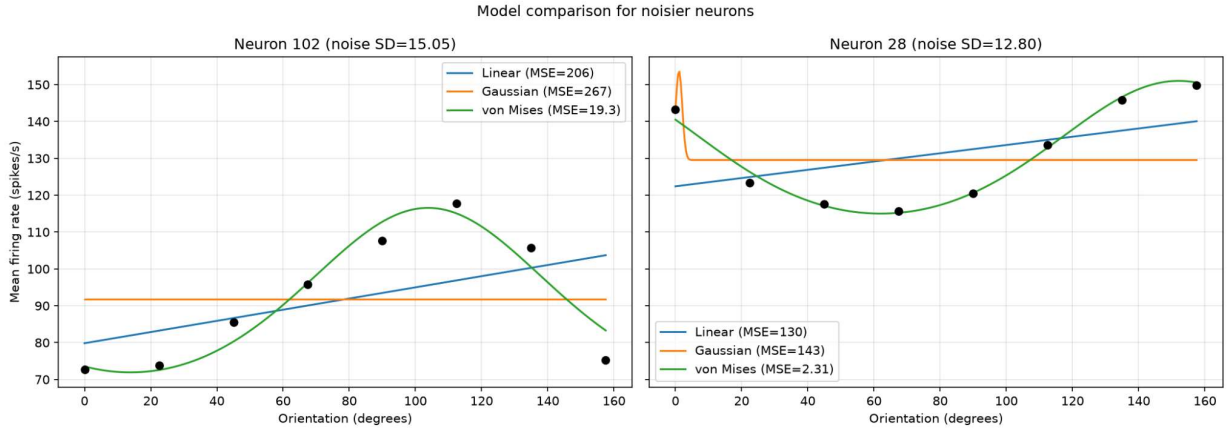
It is more appropriate for comparing neurons whose preferred orientations are near the circular boundary.

Part 5: Comparing the models and evaluating generalization

Model	Number of Parameters	MSE	Biologically interpretable?	Accounts for circularity?
Linear	2	26.359496	No	No
3rd-order polynomial	4	13.395685	No	No
4th-order polynomial	5	9.319371	No	No
7th-order polynomial	8	0.000000	No	No
Gaussian	4	4.020137	Yes	No

von Mises	4	0.067018	Yes	Yes

Select 2 more neurons whose responses are substantially noisier than those of the neuron you originally selected. For all neurons, choose only 3 different regression models, and fit each model for both of them.



Neuron	Noise standard deviation
Selected neuron 89 Noisy neuron 102 Noisy neuron 28	2.371527 15.051200 12.803825

The original selected neuron has much lower trial-to-trial variability than neurons 102 and 28.

Neuron 102

Model	MSE
Linear Gaussian von Mises	206.08951031 267.15844966 19.33085783

The von Mises model has the lowest MSE for neuron 102.

Neuron 28

Model	MSE
Linear Gaussian Von Mises	130.39117080 143.32209914 2.30524518

The von Mises model has the lowest MSE for neuron 28 as well.

Q1. Compare the goodness of fit of the different models for the two neurons.

For both noisy neurons, the linear model performs substantially worse than the nonlinear models. Comparing MSE values from the above tables, the von Mises model has the lowest MSE for both neurons. Neuron 102 has larger errors overall than neuron 28, consistent with its higher noise level

Q2. How does the increased noise affect the relative performance of the models?

Increased noise makes the tuning curve harder to estimate accurately and increases the MSE of all models. The nonlinear models still outperform the linear model because the underlying response remains better described by a tuning curve than by a straight line.

For the noisier neurons, the linear model cannot capture the tuning pattern. The Gaussian model can be affected by the circular boundary and may fit poorly. The von Mises model remains effective because it incorporates the expected circular structure of orientation.

Although noise makes parameter estimation less reliable, the structural advantage of the von Mises model remains clear in these results.

Q3. Is the model with the lowest MSE necessarily the model you would expect to generalize best?

No. The MSE measures how well a model fits the data already observed. It does not measure how well the model predicts new observations.

A highly flexible model can obtain a low training MSE by fitting random noise. This is particularly clear for the seventh-order polynomial, which has eight parameters for eight data points and achieves almost zero training error. Generalization should ideally be evaluated using a held-out test set, cross-validation or independent trials. The Gaussian and von Mises models are preferable to the seventh-order polynomial because they are more constrained and interpretable.

Q4. Which model would you select if your goal were to predict the response of the neuron to a new stimulus orientation? Explain how model complexity and noise influence your choice of model

For predicting a new response to an orientation, I would choose the von Mises model because:

- Orientation is intrinsically circular.
- The model has only four parameters, which have biological meaning.
- It avoids the oscillatory behavior of high-degree polynomials.
- It achieved the lowest MSE for both noisy neurons and for neuron 89.

The seventh-degree polynomial should not be selected solely because of its nearly zero training MSE. Its high flexibility allows it to interpolate the observed points but makes it vulnerable to overfitting and poor generalization.